

PRACTICE SHEET

1. The value of $\log 40$ is
 (1) 1.6020 (2) 1.2020
 (3) 1.94 (4) 1.3256
2. The value of $\log 8$ is equivalent to
 (1) $\log 12 + \log 4 - \log 8$
 (2) $\log 16 + \log 6 - \log 8$
 (3) $\log 6 + \log 4 - \log 2$
 (4) $\log 12 + \log 4 - \log 6$
3. Which one of the following is not correct
 (1) $\log 1 = 0$ (2) $\log_e 10 = 1$
 (3) $\log_{10} 10 = 1$ (4) $\log_{0.1} 0.1 = 1$
4. In $y = \log_e x$, value of x can never be
 (1) 2 (2) $1/2$
 (3) -1 (4) 1.6
5. The value of $\log 64$ is
 (1) 1.7658 (2) 1.8060
 (3) 1.2365 (4) 1.2546
6. If $\log 5 = k$ and $\log 125 = t$, then which of the following relation is correct?
 (1) $k = t^3$ (2) $k = 3t$
 (3) $t = 3k$ (4) $t = k^3$
7. The value of $7 \log \frac{16}{15} + 5 \log \frac{25}{24} + 3 \log \frac{81}{80}$ is :
 (1) 0.3010
 (b) 0.3512
 (3) 0.412
 (d) None of the above
8. The value of $\log \frac{70}{33} + \log \frac{22}{135} - \log \frac{7}{18}$ is:
 (given $\log 2 = 0.3010$, $\log 3 = 0.4771$)
 (a) -0.0512 (b) 0.4213
 (c) 0.3010 (d) None of these
9. $\log_{25} 625 - \log_{31} 961 + \log_{29} 841 = ?$
 (a) 2 (b) 6
 (c) 0 (d) 4
10. If $\log x + \log y = \log (x + y)$ then :
 (a) $x = y$ (b) $xy = 1$
 (c) $y = \frac{x-1}{x}$ (d) $y = \frac{x}{x-1}$
11. The value of $2^{2-\log_2 5}$ is equal to:
 (1) $4/5$ (2) $5/4$
 (3) $2/5$ (4) $5/2$
12. What is the value of $2 \log_8 2 - \frac{1}{3} \log_3 9$?
 (a) 0 (b) 1
 (c) 2 (d) $1/3$
13. What is the value of $\log_2 (\log_3 81)$?
 (a) 2 (b) 3
 (c) 4 (d) 9
14. What is $\log(a + \sqrt{a^2 + 1}) + \log\left(\frac{1}{a + \sqrt{a^2 + 1}}\right)$ equal to?
 (a) 1 (b) 0
 (c) 2 (d) $\frac{1}{2}$
15. What is the value of:
 $\log_{10} \left(\frac{9}{8}\right) - \log_{10} \left(\frac{27}{32}\right) + \log_{10} \left(\frac{3}{4}\right)$?
 (a) 3 (b) 2
 (c) 1 (d) 0
16. If $\log_3 \log_3 [\log_3 x] = \log_3 3$, then what is the value of x ?
 (a) $3a$ (b) 27
 (c) 3^9 (d) 3^{27}
17. What is the least integral value of $2 \log_{10} x - \log_x (0.01)$?
 (1) 0 (2) 2
 (3) 4 (4) 8
18. What is the value of $\log_{\sqrt{b}} a \log_{(c)^{1/3}} b \log_{(a)^{1/4}} c$?
 (1) 12 (2) 24
 (3) $1/12$ (4) $1/2$
19. The number of solutions of $\log_2 (x-1) = 2 \log_2 (x-3)$:
 (a) 2 (b) 1
 (c) 6 (d) 7
20. The value of $e^{(\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \dots + \log_{10} \tan 89^\circ)}$ is equal to:

- (a) 0 (b) 1
(c) e (d) $\frac{1}{e}$
21. What is the value of $\log_{100} 0.1$?
(1) $1/2$ (b) $-1/2$
(3) -2 (4) 2
22. The value of $\frac{1}{1+\log_c ab} + \frac{1}{1+\log_b ac} + \frac{1}{1+\log_a bc}$
(1) 0 (2) 1
(3) abc (4) $a + b + c$
23. The value of $\frac{1}{\log_{xy} xyz} + \frac{1}{\log_{yz} xyz} + \frac{1}{\log_{xz} xyz}$
(1) xyz (2) 2
(3) 0 (4) 1
24. If $\log_4 x + \log_2 x = 6$, then the value of x is ?
(1) 16 (2) 4
(3) 2 (4) 1
25. What is the value of $[\log_{10}(5\log_{10}100)]^2$?
(1) 4 (2) 3
(3) 2 (4) 1
26. If $\log_r p = 2$, $\log_r q = 3$ then the value of $\log_{pq}$ is
(1) $1/3$ (2) $2/3$
(3) $3/2$ (4) 6
27. If $\log_8 x + \log_4 x + \log_2 x = 11$, then the value of x is
(1) 128 (2) 16
(3) 32 (4) 64
28. $5^{2\log_5 4}$ is equal to
(1) 8 (2) 64
(3) 16 (4) 5
29. x triples every second. How will $\log_2 x$ change every second?
(1) It will double every second
(2) It will triple every second
(3) It increases by constant amount every second
(4) None of these
30. What is $\log_{81} 243$ equal to?
(1) 0.75 (2) 1.25
(3) 1.5 (4) 3

ANSWER KEY

1.	(1)	2.	(4)	3.	(2)	4.	(3)	5.	(2)	6.	(3)	7.	(1)	8.	(1)	9.	(1)	10.	(4)
11.	(1)	12.	(1)	13.	(1)	14.	(2)	15.	(4)	16.	(4)	17.	(3)	18.	(2)	19.	(2)	20.	(2)
21.	(2)	22.	(2)	23.	(2)	24.	(1)	25.	(4)	26.	(3)	27.	(4)	28.	(3)	29.	(3)	30.	(2)

CDS PYQ

1. The value of $\frac{1}{5}\log_{10} 3125 - 4\log_{10} 2 + \log_{10} 32$ is
[CDS 2016(I)]
(a) 0 (b) 1
(c) 2 (d) 4
2. What is the solution of $\log_{10} \left[1 - \left\{ 1 - (1 - x^2)^{-1} \right\}^{-1} \right]^{\frac{1}{2}} = 1$
[CDS 2016(II)]
(a) $x = 100$ (b) $x = 10$
(c) $x = 1$ (d) $x = 0$
3. What is the value of the equation $2^{x+2} \cdot 27^{\frac{x}{x-1}} = 9$.
[CDS 2016(II)]
(a) $2, 1 - \log\left(\frac{3}{2}\right)$ (b) $2, 1 - \log\left(\frac{2}{3}\right)$
(c) $-2, 1 - \frac{\log 3}{\log 2}$ (d) $-2, 1 - \left(\frac{\log 2}{\log 3}\right)$
4. The number of digits in 3^{30} is n and it is given that $\log_{10} 3 = 0.4771$. What is the value of n ?
[CDS 2016(II)]
(a) 13 (b) 14
(c) 15 (d) 16
5. What is the number of digits in 2^{40} ? (Given that $\log_{10} 2 = 0.3010$)
[CDS 2017(I)]
(a) 14 (b) 13
(c) 12 (d) 11
6. What is the solution of the equation $x \log_{10} \left(\frac{10}{3}\right) + \log_{10} 3 = \log_{10} (2 + 3^x) + x$?
[CDS 2017(II)]
(a) 10 (b) 3
(c) 1 (d) 0
7. If $\log_{10} 6 = 0.7782$ and $\log_{10} 8 = 0.9031$, then what is the value of $\log_{10} 8000 + \log_{10} 600$?
[CDS 2017(II)]
(a) 4.6813 (b) 5.5813
(c) 1.5813 (d) 6.6813
8. For $0 < m < 1$, which one of the following is correct?
[CDS 2018(I)]
(a) $\log_{10} m < m^2 < m < m^{-1}$
(b) $m < m^{-1} < m^2 < \log_{10} m$
(c) $\log_{10} m < m < m^{-1} < m^2$
(d) $\log_{10} m < m^{-1} < m < m^2$
9. If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, then what is the value of $\log_{100} (0.72)$?
[CDS 2018(II)]
(a) 0.9286 (b) $\bar{1}.9286$
(c) 1.8572 (d) $\bar{1}.8572$
10. There are n zeros appearing immediately after the decimal point in the value of $(0.2)^{25}$. It is given that value of $\log_{10} 2 = 0.30103$. The value of n is
[CDS 2018(II)]
(a) 25 (b) 19
(c) 18 (d) 17
11. It is given that $\log_{10} 2 = 0.301$ and $\log_{10} 3 = 0.477$. How many digits are there in $(108)^{10}$?
[CDS 2019(II)]
(a) 19 (b) 20
(c) 21 (d) 22
12. The sides of a triangle are 30cm, 28cm and 16cm respectively. In order to determine its area, the logarithm of which of the quantities are required?
[CDS 2019(II)]
(a) 37, 11, 28, 16 (b) 21, 30, 28, 7
(c) 37, 21, 11, 9 (d) 37, 21, 9, 7
13. If $\log_{10} 1995 = 3.3000$ then what is the value of $(0.001995)^{\frac{1}{8}}$?
[CDS 2019(II)]
(a) $\frac{1}{10^{0.3475}}$ (b) $\frac{1}{10^{0.3375}}$
(c) $\frac{1}{10^{0.3275}}$ (d) $\frac{1}{10^{0.3735}}$
14. Let XYZ be an equilateral triangle in which $XY = 7$ cm. If A denotes the area of the triangle, then what is the value of $\log_{10} A$? (Given that $\log_{10} 1050 = 3.0212$ and $\log_{10} 35 = 1.5441$)
[CDS 2019(II)]
(a) 4.75 cm (b) 5.50 cm
(c) 6.25 cm (d) 6.75 cm

- 15.** What is the number of digits in 7^{25} , 8^{23} and 9^{20} respectively?
[Given $\log_{10} 2 = 0.301$, $\log_{10} 3 = 0.477$, $\log_{10} 7 = 0.845$]
[CDS 2019(II)]
(a) 21, 20, 19 (b) 20, 19, 18
(c) 22, 21, 20 (d) 22, 20, 21
- 16.** How many digits are there in $(54)^{10}$? ($\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$)
[CDS 2020(I)]
(a) 16 (b) 18
(c) 19 (d) 27
- 17.** If $\log x = 1.2500$ and $y = x^{\log x}$, then what is the value of $\log y$ equal to?
[CDS 2020(I)]
(a) 40.2500 (b) 2.5625
(c) 1.5625 (d) 1.2500
- 18.** Which one of the following is the largest number among 2222^2 , 222^{22} , 22^{222} , 2^{2222} ?
[CDS 2020(I)]
(a) 2^{2222} (b) 22^{222}
(c) 222^{22} (d) 2222^2
- 19.** If the roots of quadratic equation $x^2 - 4x - \log_{10} N = 0$ are real, then what is the minimum value of N ?
[CDS 2020(II)]
(a) 1 (b) $1/10$
(c) $1/100$ (d) $1/10000$
- 20.** If $6^{3-4x} 4^{x+5} = 8$ ($\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$), then which one the following is correct?
[CDS 2020(II)]
(a) $0 < x < 1$ (b) $1 < x < 2$
(c) $2 < x < 3$ (d) $3 < x < 4$
- 21.** What is the value of $\log_{10} 31.25$ equal to?
[CDS 2021(I)]
(a) $3 - 5\log_{10} 2$ (b) $3 - 2\log_{10} 2$
(c) $5 - 5\log_{10} 2$ (d) $5 - 3\log_{10} 2$
- 22.** If $5^{x-3} = 8$, then what is x equal to?
[CDS 2021(II)]
- (a) $\frac{3}{1 - \log_{10} 2}$ (b) $\frac{3}{1 + \log_{10} 2}$
(c) $\frac{2}{1 - \log_{10} 2}$ (d) $\frac{5}{1 - \log_{10} 2}$
- 23.** If $\log_{10} x + \log_{10} x^2 = 2 \log_{10} x + 1$, then what is the value of x ?
[CDS 2021(II)]
(a) 1 (b) 2
(c) 5 (d) 10
- 24.** If the roots of the equation $x^2 - 4x - \log_{10} N = 0$ are real, then what is the minimum value of N ?
[CDS 2021(II)]
(a) 0.1 (b) 0.01
(c) 0.001 (d) 0.0001
- 25.** If $5^{x-1} = (2.5)^{\log_{10} 5}$, then what is the value of x ?
[CDS 2021(II)]
(a) 1 (b) $\log_{10} 2$
(c) $\log_{10} 5$ (d) $2\log_{10} 2$
- 26.** If $\frac{\log_{10} a}{b-c} = \frac{\log_{10} b}{c-a} = \frac{\log_{10} c}{a-b}$, ($a \neq b \neq c$), then what is the value of abc ?
[CDS 2022 (II)]
(a) -1 (b) 0
(c) 1 (d) 3
- 27.** If $\log_{10} \left[995 + \sqrt{x^2 - 12x + 60} \right] = 3$, then what is the sum of the roots of the equation?
[CDS 2022 (II)]
(a) 12 (b) 11
(c) 10 (d) 9
- 28.** What is the number of digits in the expansion of 125^{100} ? (Given $\log_{10} 2 = 0.301$)
[CDS-2023 (2)]
(a) 69 (b) 70
(c) 209 (d) 210

ANSWER KEY

1.	(b)	2.	(b)	3.	(c)	4.	(c)	5.	(b)	6.	(d)	7.	(d)	8.	(a)	9.	(b)	10.	(d)
11.	(c)	12.	(d)	13.	(b)	14.	(a)	15.	(c)	16.	(b)	17.	(c)	18.	(a)	19.	(d)	20.	(b)
21.	(a)	22.	(a)	23.	(d)	24.	(d)	25.	(d)	26.	(c)	27.	(a)	28.	(d)				